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```
clc; clear;
close all;
```

FIR FILTER PARAMETERS

```
b = [0.0625 0.2500 0.3750 0.2500 0.0625];
Fs = 1000;
N = 128;
x = [1 zeros(1,N-1)];
n = 0:N-1;
```

1. DIRECT REALIZATION

```
yd = filter(b,1,x);
```

2. CASCADE REALIZATION

```
z = roots(b);
g = b(1);
yc = g*x;
for k = 1:length(z)
    yc = filter([1 -z(k)], 1, yc);
end
yc = real(yc);
```

3. SYMMETRY AND LINEAR PHASE

```
symErr = max(abs(b - fliplr(b)));
D = (length(b) - 1) / 2;
```

```
if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end
```

4. FREQUENCY, PHASE AND GROUP DELAY

```
[H, w] = freqz(b, 1, 1024);
phase = unwrap(angle(H));
gd = grpdelay(b, 1, 1024);
```

5. NUMERICAL COMPARISON

```
err = max(abs(yd - yc));
dev = err / max(abs(yd)) * 100;
```

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DISPLAY RESULTS ==========

```
fprintf('\n==========================================\n');
fprintf('    FIR FILTER REALIZATION RESULTS\n');
fprintf('==========================================\n');
fprintf('\nFIR Coefficients:\n');
disp(b);
fprintf('Sampling Frequency = %d Hz\n', Fs);
fprintf('Number of Samples  = %d\n', N);
fprintf('Coefficient Symmetry = %s\n', symmetry);
fprintf('Symmetry Error       = %.3e\n', symErr);
fprintf('Group Delay          = %.2f samples\n', D);
fprintf('Maximum Error        = %.3e\n', err);
fprintf('Percentage Deviation = %.3e %%\n', dev);
```

```
==========================================
    FIR FILTER REALIZATION RESULTS
==========================================

FIR Coefficients:
    0.0625    0.2500    0.3750    0.2500    0.0625

Sampling Frequency = 1000 Hz
Number of Samples  = 128
Coefficient Symmetry = Symmetric
Symmetry Error       = 0.000e+00
Group Delay          = 2.00 samples
Maximum Error        = 8.327e-16
Percentage Deviation = 2.220e-13 %
```

IMPULSE RESPONSE VALUES ==========

```
fprintf('\n=========================================\n');
fprintf('FIRST 10 IMPULSE RESPONSE VALUES\n');
fprintf('=========================================\n');
fprintf('   n       Direct       Cascade\n');
for k = 1:10
    fprintf('%4d %10.6f    %10.6f\n', n(k), yd(k), yc(k));
end
```

```
=========================================
FIRST 10 IMPULSE RESPONSE VALUES
=========================================
   n       Direct       Cascade
   0   0.062500      0.062500
   1   0.250000      0.250000
   2   0.375000      0.375000
   3   0.250000      0.250000
   4   0.062500      0.062500
   5   0.000000      0.000000
   6   0.000000      0.000000
   7   0.000000      0.000000
   8   0.000000      0.000000
   9   0.000000      0.000000
```

CASCADE ZEROS ==========

```
fprintf('\n=========================================\n');
fprintf('CASCADE REALIZATION ZEROS\n');
fprintf('=========================================\n');
disp(z);
```

```
=========================================
CASCADE REALIZATION ZEROS
=========================================
  -1.0002 + 0.0000i
  -1.0000 + 0.0002i
  -1.0000 - 0.0002i
  -0.9998 + 0.0000i
```

OUTPUT PARAMETER TABLE ==========

```
Parameter = { 'Direct realization output'; ...
              'Cascade realization output'; ...
              'Impulse response'; ...
              'Phase response'; ...
              'Comparison of realizations' };
Observation = { 'Direct FIR filtering'; ...
                'Cascade of first-order sections'; ...
                'Equal to FIR coefficient sequence'; ...
                'Linear phase'; ...
                'Equivalent transfer-function behavior' };
Value = { sprintf('[%.4f %.4f %.4f %.4f %.4f]', yd(1:5)); ...
          sprintf('[%.4f %.4f %.4f %.4f %.4f]', yc(1:5)); ...
          sprintf('[%.4f %.4f %.4f %.4f %.4f]', yd(1:5)); ...
          sprintf('Group delay = %.2f samples', D); ...
          sprintf('Max error = %.3e; Deviation = %.3e %%', err, dev) };
Unit_Note = { 'Amplitude'; 'Amplitude'; 'h[n]'; 'Samples'; 'Numerical
precision' };

Output_Table = table(Parameter, Observation, Value, Unit_Note);

fprintf('\n===========================================\n');
fprintf('OUTPUT PARAMETER TABLE\n');
fprintf('===========================================\n');
disp(Output_Table);
```

```
===========================================
OUTPUT PARAMETER TABLE
===========================================
              Parameter
Observation
Value                                    Unit_Note
    ____________________________

_____________________________________________________    ____________________

    {'Direct realization output' }    {'Direct FIR
filtering'                  }    {'[0.0625 0.2500 0.3750 0.2500
0.0625]'          }    {'Amplitude'          }
    {'Cascade realization output'}    {'Cascade of first-order
sections'      }    {'[0.0625 0.2500 0.3750 0.2500 0.0625]'          }
{'Amplitude'          }
    {'Impulse response'          }    {'Equal to FIR coefficient
sequence'    }    {'[0.0625 0.2500 0.3750 0.2500 0.0625]'          }
{'h[n]'               }
    {'Phase response'            }    {'Linear
phase'                        }    {'Group delay = 2.00
```

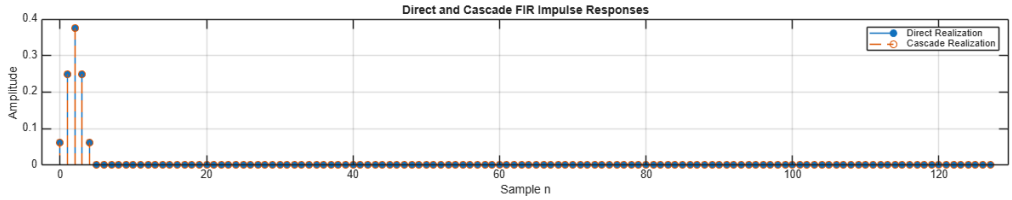
```
samples'                    }    {'Samples'           }
    {'Comparison of realizations'}    {'Equivalent transfer-function
behavior'}    {'Max error = 8.327e-16; Deviation = 2.220e-13 %'}
{'Numerical precision'}
```

==========

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GRAPH 1: DIRECT VS CASCADE IMPULSE RESPONSE
==========

```
figure;
stem(n, yd, 'filled'); hold on;
stem(n, yc, '--'); grid on;
xlabel('Sample n'); ylabel('Amplitude');
title('Direct and Cascade FIR Impulse Responses');
legend('Direct Realization', 'Cascade Realization');
```

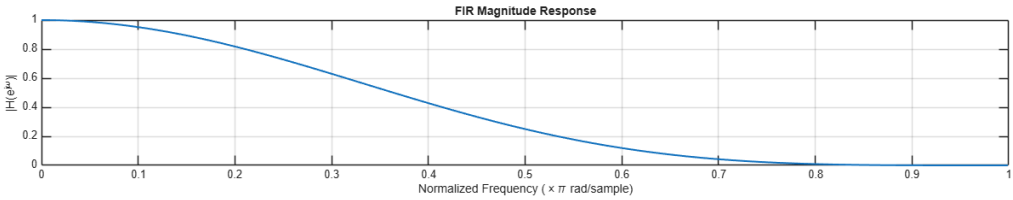


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GRAPH 2: MAGNITUDE RESPONSE
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```
figure;
plot(w/pi, abs(H), 'LineWidth', 1.5); grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)'); ylabel('|
H(e^{j\omega})|');
title('FIR Magnitude Response');
```

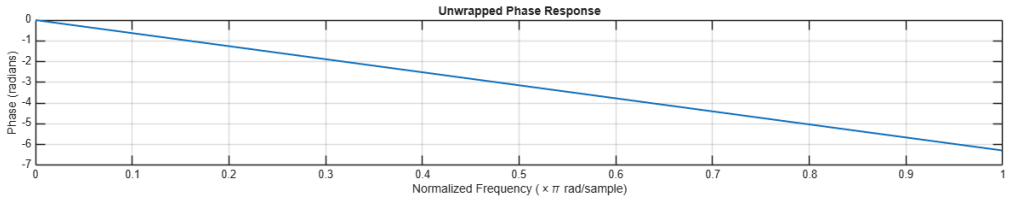


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GRAPH 3: PHASE RESPONSE ==========

```
figure;
plot(w/pi, phase, 'LineWidth', 1.5); grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)'); ylabel('Phase
(radians)');
title('Unwrapped Phase Response');
```

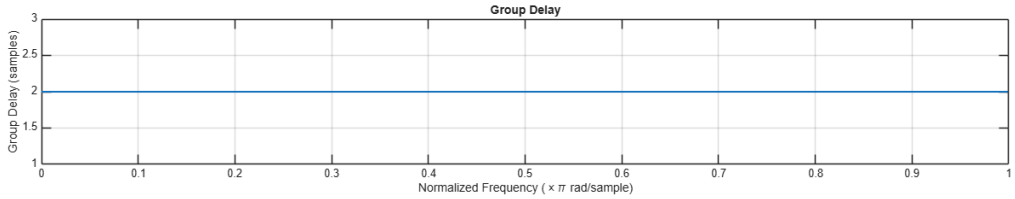


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GRAPH 4: GROUP DELAY ==========

```
figure;
plot(w/pi, gd, 'LineWidth', 1.5); grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)'); ylabel('Group Delay
(samples)');
title('Group Delay');
```

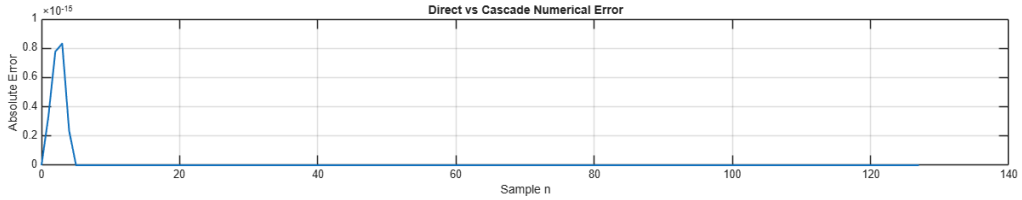


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GRAPH 5: NUMERICAL ERROR ==========

```
figure;
plot(n, abs(yd - yc), 'LineWidth', 1.5); grid on;
xlabel('Sample n'); ylabel('Absolute Error');
title('Direct vs Cascade Numerical Error');
```



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FINAL RESULT ==========

```
fprintf('\n=========================================\n');
fprintf('FINAL RESULT\n');
fprintf('=========================================\n');
fprintf('1. FIR filter realized using Direct structure.\n');
fprintf('2. FIR filter realized using Cascade structure.\n');
fprintf('3. Coefficients are %s.\n', symmetry);
fprintf('4. Linear-phase group delay = %.2f samples.\n', D);
fprintf('5. Maximum realization error = %.3e.\n', err);
fprintf('6. Percentage deviation = %.3e %%.\n', dev);
fprintf('7. Direct and Cascade responses are equivalent.\n');
```

```
=========================================
FINAL RESULT
=========================================
1. FIR filter realized using Direct structure.
2. FIR filter realized using Cascade structure.
3. Coefficients are Symmetric.
4. Linear-phase group delay = 2.00 samples.
5. Maximum realization error = 8.327e-16.
6. Percentage deviation = 2.220e-13 %.
7. Direct and Cascade responses are equivalent.
```

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